

Stat 150 – Lecture 02

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1 Markov Chain 2.0

1.1 Classification of States:

Understand the behavior of Markov chains. Sometimes they are recurrent, sometimes transient, etc.

Notation $i \rightarrow j$ denotes j is reachable or "accessible" from i .

If there is a positive probability that in finite steps the Markov Chain starts at i and ends at j . We can denote this

$\exists n \in \mathbb{Z}^+$ such that $P_{ij}^n =$ probability of jump to j at n steps starting at $i > 0$

Definition 1.1 (Chapman-Kolmogorov Equation - Discrete Case). Given $i \leftrightarrow j$ and $j \leftrightarrow k$ to find reachability between i and k . Basically a matrix product to consider reachability between i and k by considering an intermediate in some amount of steps.

$$P_{ik}^{m+n} = \sum P_{ir}^m P_{rk}^n$$

Notation:

1. $\mathbb{P}_x(A) = \mathbb{P}(A \mid x_0 = x)$ for an event A in a markov chain $(x_n)_{n \in \mathbb{N}}$
2. Denote $T_y = \min\{n \geq 1 : X_n = y\}$ time of the first return to y . (y at time 0, return back).
- 3.

$$\mathbb{P}_{yy} = \mathbb{P}_y(T_y < \infty) = \sum_{n=1}^{\infty} \mathbb{P}_y(T_y = n)$$

denotes probability if we start at y we return in a finite number of steps.

Definition 1.2 (Strong Markov Property). Formalizes the memoryless nature of Markov Chains to show results like k th return is given by $(P_{yy})^k$ by this property.

Definition 1.3 (Stopping Time). T is a stopping time if the occurrence/non-occurrence of event $\tau = n$ i.e "We stop at time T " and determined by some of $X_0, X_1, \dots, X_n$.

1.2 Durrett: Strong Markov Chain Properties Let $(X_t)_{t \in \mathbb{N}}$ be a Markov Chain.

Let τ be a stopping time, and E an event prior to τ (relevant to $X_0, X_1, \dots, X_n$)

Then, we have

$$\forall i, j \in S, \mathbb{P}(X_{\tau+k} = j \mid X_\tau = i, E) = \mathbb{P}(X_{\tau+k} = j \mid X_\tau = i)$$

If you define $(Y_n) := x_{\tau+n}, \forall n \geq 0$ is a markov chain.

Sketch of Proof: We have to eventually show that

$$\mathbb{P}(X_{\tau+1} = z \mid X_\tau = y, \tau = n) = P_{yz} = P(y, z)$$

2 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

3 Examples and calculations

Example 3.1 (Lecture example). Write the setup, the calculation, and the conclusion.

4 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: